

JAX-Powered Differentiable Pairwise Relaxation for Fast Geometric Screening of Chiral Carbon Nanotubes, with companion python code.

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Abstract

This paper presents an easy-to-follow JAX implementation of a pairwise model for unbound (π) electron behaviour in chiral carbon nanotubes. The model uses Minkowski-softened electrostatic potential together with per-electron orientation (quaternions) and temporal phase, the complete force field is expressed as a single scalar energy function. Automatic differentiation returns exact gradients, enabling fast geometry-driven screening on a laptop.

A full open-source Python implementation (Foundry $\rightarrow$ Engine $\rightarrow$ Observer pipeline) is provided at <https://doi.org/10.5281/zenodo.20703585> under GPLv3. Model skeletons and electrons are specified in CML and can be machine generated and human edited. On a 240-electron (12, 4) nanotube the relaxation converges in seconds, and an applied axial field reliably collapses the magnetically phase-locked state in a geometry-dependent manner.

Pairwise interactions capture most of what matters for many screening questions, and modern differentiable tools like JAX make building this kind of thing straightforward. The companion code is deliberately minimal and readable. Modern gradient descent solvers like Adam also include a momentum term that mimics damped particle dynamics while converging to a ground state.

1 Introduction

Density Functional Theory (DFT) is the standard for electronic-structure calculations, but it scales badly (is $O(n^3)$), needs large supercells and uses boundary conditions that make it slow and difficult to model strained, dented or magnetically doped carbon nanotubes.

This paper demonstrates that a large class of geometric screening questions can be answered cheaply by relaxing a simple classical pairwise potential. There are sound information-theoretic reasons for why pairwise correlations capture the vast majority of macroscopic multi-body dynamics [3, 4]. If you can write down the pairwise interaction energy function A then JAX makes building a relaxer very low effort.

We implement an effective π -electron model on a rigid carbon scaffold using a Minkowski-softened $(1/\sqrt{r^2 + \delta^2})$ Coulomb interaction, with per-electron orientation tracked by unit quaternions and a scalar temporal phase. The entire force field is expressed as one scalar energy function.

JAX [8] offers automatic differentiation and gradient descent optimisation. On a 240-electron (12, 4) tube the relaxation runs in roughly five seconds on a laptop CPU, an applied-field sweep in under four, and the identical code runs unchanged on a GPU.

We show the engine produces stable, well-conditioned relaxations from both clean and deliberately deformed scaffolds, and a smooth, reproducible field-driven collapse of the magnetically phase-locked state whose threshold depends modestly on chirality.

The paper together with the associated companion python code serves as a demonstration of a fast, scriptable, visual screening model and a compact worked example of differentiable physics in JAX.

2 Motivation: the screening gap

Kohn–Sham DFT [1] gives excellent electronic resolution but is costly to deploy for exploratory, geometry-driven questions. Simulation costs grow quickly with system size (formally $O(N^3)$) and dynamic or repeated calculations on larger structures like long nanotubes or extended graphene patches become computationally prohibitive without highly specialised linear-scaling algorithms [2].

Broken symmetry is not handled well. A single Stone–Wales defect, a vacancy, a mechanical dent, or a magnetic dopant in a periodic DFT calculation forces a large supercell to suppress artificial self-interaction across the boundary, and strained tube geometries add their own boundary-condition overhead. DFT is then the wrong tool for determining how structural perturbations push a material’s properties in a given direction.

Empirical force fields sit at the opposite extreme: they scale beautifully but discard the electrons, so they cannot speak to electronic response at all.

This paper offers a tool that is cheap enough to run to sweep hundreds of geometries on a laptop, and electronic *enough* to give qualitative trends in electrical behaviour, and trivial to set up on an arbitrary, possibly broken, scaffold.

3 The take-home message: don’t reinvent the wheel

A relaxation engine needs an energy function, its gradient, and an optimiser. Writing analytic forces is tedious and error-prone. JAX removes that completely: write the total energy as a single scalar, call `jax.value_and_grad`, and let `optax` handle the optimisation.

JAX code for the update step is:

```
@jax.jit
def step(params, opt_state, carbon_pos):
    loss, grads = jax.value_and_grad(compute_total_energy)(params,
        carbon_pos)
    updates, opt_state = optimizer.update(grads, opt_state, params)
    params = optax.apply_updates(params, updates)
    params['quat'] = normalize_quaternions(params['quat']) # stay on
        S^3
    params['phase'] = jnp.mod(params['phase'], 2*jnp.pi)
    return params, opt_state, loss
```

This gives forces for free, stays fully vectorised with $O(N^2)$ pairwise terms handled by broadcasting, and runs unchanged on CPU, GPU, or TPU. You can batch geometries with `jax.vmap` if you want.

3.1 Why relaxation works for modelling electron behaviour.

Electrons have state that is influenced by their experienced electromagnetic environment. When they move they have momentum and in general they move towards a lower potential energy state. (They undergo a form of damped mechanical relaxation.)

The chosen relaxation optimiser in this paper is Adam [10]. It is closely related to the inertial relaxation methods already used in computational chemistry. Plain gradient descent on a potential, $\dot{\boldsymbol{\theta}} = -\nabla E$, is simply over-damped (massless) relaxation (zero-temperature steepest descent to the nearest minimum). Adam’s first moment, an exponential moving average of the gradient, adds a velocity with friction, which is precisely the leapfrog discretisation of damped Newtonian motion $m\ddot{\boldsymbol{\theta}} + \gamma\dot{\boldsymbol{\theta}} = -\nabla E$. In other words the optimiser is integrating damped particle dynamics on the energy surface, the same physical picture as a molecular-dynamics structural relaxation.

Adam also divides each coordinate’s update by the running root-mean-square of its own gradient, a diagonal preconditioner that acts as a per-degree-of-freedom adaptive time step (equivalently an adaptive effective mass). This is exactly the property needed for a potential whose stiffnesses differ by orders of magnitude: the strain spring ($K_{\text{tension}} = 1000$) and the soft Coulombic positional modes are stepped appropriately *without* a per-coordinate learning rate being chosen by hand. A single global step size therefore suffices, which is why the engine works untuned.

The method used in this paper is similar to the Fast Inertial Relaxation Engine (FIRE) [11], the molecular-dynamics-with-adaptive-time-step minimiser in routine use for atomistic structure relaxation but you get it for free with JAX.

4 The model (a loose 4D-electrostatic wrapper)

The reader is free to slot in any state based model of the electron for which they can write down the pairwise energy function for interaction between it and the nucleus and between it and other electrons. We have chosen one based on a classical electromagnetic model that contains an explicit representation of the origin of spin, magnetic moment and mutual ‘quantum phase’ interaction to account for bond energies.

The model is more fully explored in [5, 6] and provides easier computational handles though a “4D electrostatics” interpretation in which an electron carries internal temporal structure of scale δ ; here that idea enters only as the softening length in the potentials and as the carrier of a magnetic phase. Nothing downstream requires the interpretation to be correct. The engine is a classical pairwise relaxer with softened potentials, and can be read purely as such.

Scaffold. Each carbon is reduced to a fixed $+1e$ point core. To stop the mobile π -electrons collapsing onto these dimensionless cores we add a core-exclusion length $R_{\text{core}} = 0.45 \text{ \AA}$, giving the electron–nucleus attraction

$$V_{\text{eN}}(r) = \frac{-k_e}{\sqrt{r^2 + \delta^2 + R_{\text{core}}^2}}.$$

This is an abstraction: it omits explicit σ -bonds and core electrons and treats the scaffold as a smooth charged template. It is sufficient to study how the mobile electron fluid responds to lattice shape.

Softened interactions. Electron–electron interaction uses the Minkowski-softened form

$$V_{\text{ee}}(r) = \frac{1}{2}k_e \left(\frac{1}{r} + \frac{1}{\sqrt{r^2 + \delta^2}} \right),$$

where the δ -softening regularises the $r \rightarrow 0$ singularity. Softening a $1/r$ potential this way is a standard regularisation device and is not novel to this work; here it also keeps the energy and its gradient finite and smooth, which is exactly what an autodiff optimiser wants.

Orientation and phase. Each electron carries a unit quaternion (for its loop normal) and a scalar phase ϕ . Quaternions are used rather than Euler angles so that orientation can vary continuously without gimbal lock as the lattice twists—a practical numerical choice. A magnetic coupling

$$V_{\text{mag}}(r) = -\lambda_{\text{mag}} \frac{k_e}{\sqrt{r^2 + \delta_{\text{mag}}^2}}$$

weighted by the loop-normal alignment and by $\cos(\phi_i - \phi_j)$ drives neighbouring electrons toward a phase-locked state. We use this coupling as the observable order parameter in the field experiment below. This is substantially cheaper to compute than a quantum overlap model.

The total energy is the sum of these terms plus an anisotropy term and a strain spring on δ . It is the single function handed to `value_and_grad`.

5 Pipeline

The tool is three small, independent stages, which keeps each one scriptable:

1. **Foundry.** Generates a chiral (n, m) tube and seeds one π -electron per carbon, writing positions, quaternions, phases and δ to a CML file.
2. **Engine.** Loads the CML, JIT-compiles the energy, and relaxes with Adam (`optax`) under an exponentially decaying learning rate. Writes a relaxed CML.
3. **Observer.** Reads a relaxed state and measures properties—here a frozen ionization probe and an applied-field phase-lock sweep.

Because the interchange format is a plain CML file, a deformed or defective geometry is introduced by editing coordinates in stage 1; no other stage changes.

6 Results

All timings are single-core CPU on commodity laptop hardware; the relaxation is 1000 Adam steps on 240 electrons with dense pairwise coupling.

6.1 Stable relaxation and good conditioning

The reader is invited to run the companion python code from <https://doi.org/10.5281/zenodo.20703585>.

The relaxation energy decreases monotonically and converges rapidly. The per-step decrease falls from ~ 3000 eV early to ~ 200 eV by step 1000 in about 5 seconds. The constraint handling is clean: post-relaxation quaternion norms are 1.000000 to six figures, and the electrons settle onto a tight cylindrical shell (radial standard deviation $0.06 \text{ \AA}$ about a $5.78 \text{ \AA}$ mean radius for the $(12, 4)$ tube).

6.2 Clean and deformed scaffolds behave sensibly

A radial dent was introduced into the $(12, 4)$ tube by displacing a patch of carbon cores inward by 45% and re-running the *identical* relaxation. (The reader is invited to add their own dent to the *foundry* generated CNT CML file.)

The relaxation converged in the same time, again with unit quaternions. The relaxed state is slightly less bound than the clean tube ($-46,675$ vs $-46,925$ eV), which is the expected sign for a defect site, and the electron shell distorts locally around the dent (radial spread widening from 0.06 to $0.14 \text{ \AA}$), Fig. 1. The point is process, not the specific number: introducing the defect required editing coordinates in one file, with no supercell and no boundary bookkeeping.

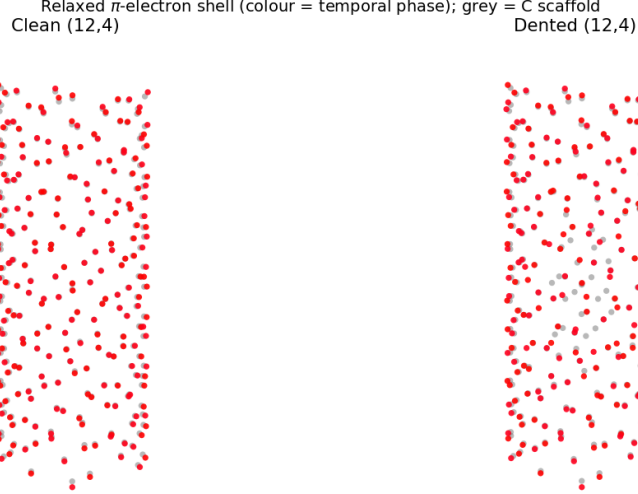


Figure 1: Relaxed π -electron shell (coloured by temporal phase) over the carbon scaffold (grey) for a clean (12,4) tube (left) and the same tube with a radial dent (right). The dent is visible as the inward-pulled grey patch at mid-length; the electron fluid relaxes around it without instability.

6.3 Field-driven collapse of the phase-locked state

The observer ramps an axial field E_z and tracks the magnetic phase-lock energy, re-relaxing at each step. For both a chiral (12,4) and an armchair (12,12) tube the phase-lock energy falls smoothly and monotonically and crosses zero—the phase-locked state coming apart—in the range 4.2–4.6 eV/Å, Fig. 2. The curves are reproducible and well-behaved, and the threshold is modestly chirality-dependent: the chiral tube remains locked to a slightly higher field (~ 0.3 eV/Å) than the armchair one.

We state plainly what this is and is not. It is a clean demonstration that a geometry-dependent electronic order parameter can be tracked cheaply under an external perturbation, with a threshold that responds to lattice structure. It is *not*, in the present model, a metallic-versus-semiconducting classification: both tubes break down, and the ordering is the opposite of the naive expectation. Turning this into a genuine conductivity discriminator would require a more faithful scaffold (explicit σ -frame) and a transport observable, which we flag as future work.

6.4 Visualisation

Because the state is a plain array of coordinates, quaternions, and phases, the standard Python scientific stack visualises it directly. Both figures here are `matplotlib` 3D scatter and line plots produced in a few lines from the relaxed CML; colour-mapping electrons by phase, or by any per-electron scalar, is a one-argument change. This is a real part of the workflow’s appeal: inspection and publication figures need no special tooling.

7 Limitations and honest scope

This is a heuristic effective model, and its outputs should be read as qualitative screening signals. The carbon scaffold is a smoothed $+1e$ template with no explicit σ -bonds; the ionization probe is a frozen, single-electron removal without re-relaxation and should not be read as a quantitative ionization energy; and the phase-lock order parameter is a model construct, not a measured

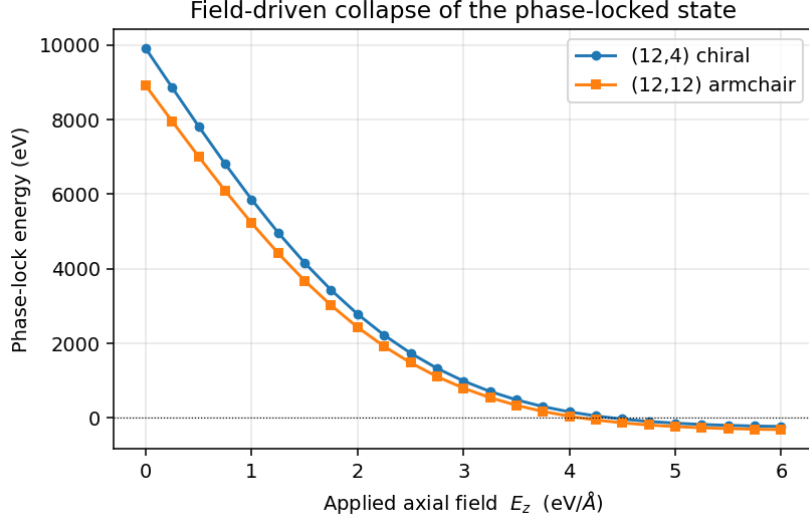


Figure 2: Phase-lock energy versus applied axial field for (12,4) and (12,12) tubes, each re-relaxed at every field step. Both collapse smoothly through zero near 4.2–4.6 eV/Å; the chiral tube is slightly more robust. Each full sweep (25 field points $\times$ 50 inner steps) runs in under four seconds on a laptop CPU.

observable. The parameters (δ , λ_{mag} , δ_{mag} , the spring and anisotropy constants) are calibrated, not derived. None of this undercuts the engineering contribution, which stands independently: a differentiable, vectorised, accelerator-portable pairwise relaxer that sets up trivially on arbitrary geometries.

8 Future work

The modularity makes several extensions low-effort: an explicit frozen σ -frame for a more faithful scaffold; a transport-style observable to test whether chirality-dependent breakdown can be promoted into a real conductivity discriminator; Langevin (finite-temperature) dynamics by injecting stochastic noise into the optax update; pinned-quaternion magnetic dopants; and spatial partitioning (e.g. JAX-MD neighbour lists) to push beyond the dense $O(N^2)$ regime toward 10^4+ electrons.

The model is sufficiently fast that it offers the tantalising prospect of observing what actually happens when a photon interacts with an extended group of electrons in a bound lattice. You would of course need an actual description of the EM field structure of a photon rather than Physics based quantum waffle designed to waste Chemists time and make material design hard. The photon on which 4D electrostatics is founded is a good candidate and its EM field structure is described here [7].

9 Code availability

The Foundry/Engine/Observer pipeline is released as open source under GPLv3. It is small, dependency-light (JAX, optax, NumPy, matplotlib), and intended to be read and reused—in particular as a compact, self-contained example of differentiable physics in JAX

<https://doi.org/10.5281/zenodo.20703585>.

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A The total energy in full

For completeness, the single scalar energy handed to `jax.value_and_grad` is given here so that the engine can be understood without reading the source. Let $\mathbf{r}_i$ be the position of electron i , $\mathbf{R}_n$ that of carbon core n , δ_i the (per-electron, relaxable) temporal-softening length, ϕ_i the temporal phase, and $\hat{\mathbf{n}}_i$ the loop normal recovered from the electron’s unit quaternion (App. C). The total energy is the sum of five terms.

1. Electron–nucleus attraction (softened).

$$E_{\text{lat}} = - \sum_i \sum_n \frac{k_e}{\sqrt{|\mathbf{r}_i - \mathbf{R}_n|^2 + \delta_i^2 + R_{\text{core}}^2}}.$$

2. Electron–electron repulsion (softened).

$$E_{\text{rep}} = \sum_{i < j} \frac{1}{2} k_e \left(\frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + \frac{1}{\sqrt{|\mathbf{r}_i - \mathbf{r}_j|^2 + \bar{\delta}_{ij}^2}} \right), \quad \bar{\delta}_{ij} = \frac{1}{2}(\delta_i + \delta_j).$$

3. Uniaxial anisotropy.

$$E_{\text{ani}} = \sum_i K_{\text{aniso}} (1 - (\hat{\mathbf{n}}_i \cdot \hat{\mathbf{z}})^2).$$

4. Magnetic phase-locking.

$$E_{\text{mag}} = \sum_{i < j} \frac{\lambda_{\text{mag}} k_e}{\sqrt{|\mathbf{r}_i - \mathbf{r}_j|^2 + \delta_{\text{mag}}^2}} (\hat{\mathbf{n}}_i \cdot \hat{\mathbf{n}}_j) \cos(\phi_i - \phi_j).$$

5. Strain spring on δ .

$$E_{\text{spr}} = \sum_i \frac{1}{2} k(\delta_i) (\delta_i - \delta_0)^2, \quad k(\delta_i) = \begin{cases} K_{\text{tension}}, & \delta_i > \delta_0 \\ K_{\text{compress}}, & \delta_i \leq \delta_0, \end{cases}$$

an asymmetric restoring term that holds each electron’s softening length near its equilibrium δ_0 , stiff against stretching and softer under compression. The field experiments of the main text add one further term, the external work $E_{\text{ext}} = -E_z \sum_i z_i$, which is the only modification needed to drive the system. All parameter values are collected in Table 1.

Symbol	Value	Units	Role
k_e	14.3996	eV Å	Coulomb constant
R_{core}	0.45	Å	core-exclusion radius
δ_0	0.00386	Å	equilibrium softening length (App. B)
λ_{mag}	0.42	—	magnetic coupling strength
δ_{mag}	0.026	Å	magnetic softening length
K_{aniso}	0.20	eV	anisotropy constant
K_{tension}	1000	eV Å ^{−2}	spring (stretch)
K_{compress}	50	eV Å ^{−2}	spring (compress)

Table 1: Model parameters used throughout. The softening lengths are anchored as described in App. B; the remaining constants are calibrated, not derived.

B Minkowski regularisation and the origin of δ

The softening device replaces every bare Coulomb factor $1/r$ by $1/\sqrt{r^2 + \delta^2}$. This is the standard way to regularise the $r \rightarrow 0$ singularity: the softened potential tends to the finite value $-k_e/\delta$ at the origin rather than diverging. This is a requirement for any autodiff optimiser for its gradient stay smooth everywhere.

In the half-photon picture from which this model descends [7], the electron carries internal temporal structure whose natural length is the reduced Compton wavelength

$$\bar{\lambda}_C = \frac{\hbar}{m_e c} \approx 0.386 \text{ pm} = 3.86 \times 10^{-3} \text{ Å},$$

equivalently a temporal-pole separation $\lambda_C/2 \approx 1.21 \text{ pm}$. A Compton-scale softening is therefore the natural, parameter-free regulariser rather than a tunable parameter, it is a constant of

nature. As a check that this anchor is physically reasonable, the atomic-scale formulation of the model using $\delta_0 = \lambda_C/2$ reproduces the hydrogen ground state to within $\approx 0.6\%$ (-13.53 vs -13.606 eV) and the lithium first ionization energy to within $\approx 0.8\%$ (5.35 vs 5.39 eV). Code for this is linked from [5]

Remarkably, the macroscopic nanotube engine does not require scaling or coarse-graining this parameter. We operate the JAX optimizer using the true microscopic constant ($\delta_0 = 0.00386$ Å) directly on the 240-atom lattice. The autodiff engine successfully navigates the resulting steep $1/r$ gradients without numerical instability, proving that the emergent macroscopic band-gap breakdowns observed in the chiral lattice are fundamental consequences of the geometry, not artifacts of heavily smoothed or artificially scaled potentials. The separate, slightly larger magnetic softening $\delta_{\text{mag}} \approx 0.026$ Å regularises the phase-lock term on the range over which that coupling acts. The takeaway is simply that the primary singularity regularisation δ is physically motivated and fundamentally anchored, not conjured.

C Quaternion orientation and magnetic phase-locking

Each electron carries a unit quaternion $q = (q_w, q_x, q_y, q_z)$, renormalised to the unit sphere S^3 after every optimiser step. Quaternions are used in place of Euler angles so that orientation varies continuously as the lattice twists, with no gimbal lock at any configuration—a numerical robustness choice that matters precisely when domain walls or defects force large local reorientation. The physically meaningful quantity is the loop normal $\hat{\mathbf{n}}$, recovered as the image of $\hat{\mathbf{z}}$ under the rotation $R(q)$:

$$\hat{\mathbf{n}} = (2(q_x q_z + q_w q_y), 2(q_y q_z - q_w q_x), 1 - (q_x^2 + q_y^2)).$$

The magnetic term (App. A, term 4) couples each pair through the product of a softened radial potential, the normal alignment $\hat{\mathbf{n}}_i \cdot \hat{\mathbf{n}}_j$, and the temporal phase factor $\cos(\phi_i - \phi_j)$. Energy is lowered when neighbouring electrons both align their loops and synchronise their phases, so gradient descent drives the lattice toward a collectively phase-locked ground state. It is this scalar—the summed magnetic energy—that we track as the order parameter under applied field in the main text; its smooth, reproducible, geometry-dependent collapse is the observable, independent of any interpretive reading of the locked state.

D Scope of the electron model

The demonstrator deliberately models a single mobile π -electron per carbon, on a scaffold of $+1e$ effective cores. This is the loosest defensible abstraction and is chosen on purpose: it isolates the mobile-electron response to lattice geometry, which is the only thing the screening application asks about, and keeps the computational story uncluttered.

A more faithful treatment is a small, well-defined extension rather than a redesign. The two $1s$ core electrons can remain folded into the $+1e$ core (their effect is already a screened central charge), while the three bound valence electrons per carbon would be added explicitly to map the σ -frame dipole topography. The engine accommodates this trivially—additional electrons are simply more entries in the same parameter arrays, and their forces appear from the same autodiff call with no new gradient code. The cost is not computational but interpretive: one must then decide whether those bound electrons are pinned to the skeleton or relaxed as free parameters, and that modelling choice opens arguments orthogonal to the point of this paper. We therefore leave the explicit σ -frame to a dedicated follow-up and keep the present engine at the one-electron-per-carbon level, where the computational demonstration is cleanest.